Total No. of Questions : 10]

SEAT No. :

P3669

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[5561]-302

B.E. (Information Technology)

MACHINE LEARNING

(2012 Course) (Semester - I) (End Semester) (414455)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Neat diagrams must be drawn wherever necessary.*
2) *Figures to the right side indicate full marks.*
3) *Use of Calculator is allowed.*
4) *Assume suitable data if necessary.*

Q1) a) Define and give example of predictive and descriptive models. **[4]**

b) Explain and differentiate Supervised and Unsupervised learning with one example each. **[6]**

OR

Q2) a) How cross validation method improves the overall performance of the machine learning model? **[6]**

b) Define following w.r.t. Binary Classification **[4]**

i) True positive. ii) False positive.

iii) True negative. iv) False negative.

Q3) a) Define with respect to Classification : **[4]**

i) Precision. ii) Recall.

iii) F1-Measure. iv) True Negative Rate.

b) With the mathematical model define Squared error and Mean-Squared Error. **[6]**

OR

Q4) a) Define regression model with one example. **[4]**

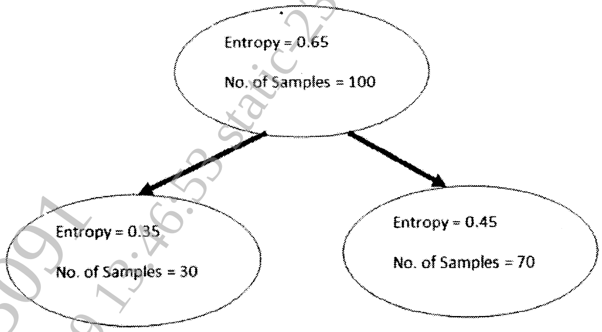
b) Define and explain : **[6]**

i) Soft-margin SVM.

ii) Slack Variable.

P.T.O.

Q5) a) Calculate the value of the information gain in the following example. **[6]**



b) There are six clusters - A, B, C, D, E and F. Following 6 × 6 distance matrix represents distances between these clusters. **[12]**

Solve it using Hierarchical Clustering algorithm and Single Linkage Method and draw the associated dendogram of it.

	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

OR

Q6) a) Find all association rules in the following database with minimum support = 2 and minimum confidence = 50%. **[8]**

TID	Items
1	A, B, E
2	B, D
3	B, C
4	A, B, D
5	A, C
6	B, C
7	A, C
8	A, B, C, E
9	A, B, C

b) Ten visitors visited the meusum. Following are their ages. We want to group them into two different groups according to their ages. Initial value of the cluster is decided as $C_1 = 16$ and $C_2 = 22$. Solve it using k-means clustering algorithm. Consider Euclidean distance measure. **[10]**

Ages of the visitors are : 15, 15, 16, 19, 22, 28, 35, 40, 60, 65.

Q7) a) Differentiate between generative and descriptive Probabilistic models with suitable examples. **[8]**

b) You are given following set of training examples. Each feature can take on one of the three nominal values : a, b or c. **[8]**

A1	A2	A3	Class
a	c	a	+
c	a	c	+
a	a	c	–
b	c	a	–
c	c	b	–

If A1 = a, A2 = c and A3 = b, then how would a Naïve Bayes System Classify the example.

OR

Q8) a) Write short note on Normal or Gaussian Distribution. **[8]**

b) Define and compare Logistic Regression with Linear Regression. **[8]**

Q9) a) What are the applications of Deep Learning? **[8]**

b) Explain 'Random Forest Algorithm' in detail. **[8]**

OR

Q10) a) How active learning can perform data stream classification? **[8]**

b) What is "Reinforcement Learning". Comment on "Giving pentalies and awards in reinforcement learning is good or not"? **[8]**

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